

Bolla Lokesh Reddy

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EDUCATION

Indian Institute of Technology Hyderabad

2023 – 2027

B.Tech in Computer Science and Engineering

CGPA: 8.3/10

EXPERIENCE

Adobe – Content Intelligence Team

May 2026 – Jul 2026

Machine Learning Engineering Intern

India

- Developed an end-to-end email layout featurization pipeline combining DOM geometry with multimodal semantic representations for structural analysis.
- Designed deterministic extraction of 22 structural features and evaluated DINOv2/Gemini embeddings with clustering methods to discover recurring layout families.
- Validated the pipeline across two independent datasets, producing coherent clusters and interpretable structural representations.

PROJECTS

Cross-Sectional Momentum Alpha Research & Backtesting

Python, NumPy, Pandas, yfinance

- Researched a 12–1 month cross-sectional momentum signal using a dollar-neutral long/short portfolio with monthly rebalancing across liquid U.S. equities.
- Built a vectorized backtesting framework with point-in-time alignment, transaction costs, slippage, out-of-sample evaluation, and automated safeguards against look-ahead bias.

Fault-Tolerant Distributed Job Processing System

Python, FastAPI, Redis Streams, PostgreSQL

- Built a fault-tolerant asynchronous job processing platform with worker pools, persistent job state, retries, dead-letter queues, and crash recovery over Redis Streams.
- Implemented at-least-once delivery with stale-message reclamation, exponential-backoff retries, and hash-based idempotency to make repeated job execution safe.

Distributed Bitcoin Mining Framework

C++, MPI, Pthreads, OpenSSL

- Designed parallel nonce-search implementations using static multithreading, dynamic work stealing, and MPI-based distributed execution with SHA-256 hashing.
- Implemented dynamic work distribution, Raft-based leader election, checkpointing, and fault recovery while analyzing scalability, load balancing, and communication overhead.

Hybrid Search & Retrieval Engine

Python, FAISS, Sentence Transformers

- Built a retrieval engine combining a from-scratch inverted index and BM25 scoring with dense Sentence Transformer embeddings and FAISS vector search.
- Implemented Reciprocal Rank Fusion and a labeled evaluation framework to compare lexical and semantic retrieval strategies and analyze their tradeoffs.

SCHOLASTIC ACHIEVEMENTS

JEE Advanced 2023 AIR 953, Top 0.5%

JEE Mains 2023 AIR 1541, Top 0.15%

COMPETITIVE PROGRAMMING

Codeforces 1400+ rating 

LeetCode 500+ problems solved 

TECHNICAL SKILLS

Languages: C++, Python, C, SQL

Quantitative: Probability, Statistics, Time-Series, Backtesting

Numerical: NumPy, Pandas

Systems: MPI, Pthreads, Redis, PostgreSQL

ML/IR: PyTorch, NLP, BM25, FAISS

Tools: Git, GitHub, Linux, CMake, Catch2

RELEVANT COURSEWORK

Computer Science: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, Distributed Computing, Computer Architecture, Compiler Design

Mathematics: Probability and Random Variables, Calculus, Discrete Mathematics